

# Chapter 95

# Composite Nodes

This chapter details the Dissolve and Merge nodes available in Fusion.

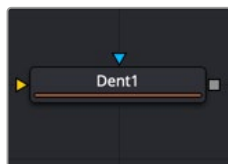
The abbreviations next to each node name can be used in the Select Tool dialog when searching for tools and in scripting references.

For purposes of this document, node trees showing MediaIn nodes in DaVinci Resolve are interchangeable with Loader nodes in Fusion Studio, unless otherwise noted.

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# Dissolve [Dx]



The Dissolve node

## Dissolve Node Introduction

The Dissolve node is typically used to mix two images together, providing a gradual transition between two clips. A Background/Foreground slider controls the mix between the foreground and background images. Dissolves are commonly used to transition between one clip and another and are a very common effect in editing. However, you can also use the extreme left and right positions of the Background/Foreground slider to switch between inputs. Unlike other nodes in Fusion, the Dissolve node does not require you to connect an image to the background but lets you output either the background or foreground according to the setting of the Background/Foreground slider.

This quality makes it possible for you use the Dissolve node as an automatic layer switching tool when connected to background and foreground clips with different durations. Simply connect each clip to the background and foreground inputs, respectively, and set the Background/Foreground slider to the input of shorter duration, to determine which is “on top.” After the last frame of that clip has ended, the Dissolve node automatically switches to the clip that’s connected to the other input.

Besides the default dissolve, the Gradient Wipe setting of the Operation menu allows you to create arbitrary animated dissolve patterns based on the luminance of an image connected to the optional Gradient Wipe input. You can use this capability with images of geometric shapes or gradients of different kinds, movie clips of fire, water ripples, or rain, the Fast Noise node, or even particle systems you create within the Fusion page to create a variety of unique and creative transitions. Soft-edged effect masks may also be used to add to the possible effects.

Ultimately, animating the Background/Foreground control allows you to control the transition that’s being used to switch from the foreground input to the background, or vice versa.

## Inputs

The Dissolve node provides three image inputs, all of which are optional:

**Background:** The first of two images you want to switch between or mix. Unlike most other nodes, it is unnecessary to connect the background input before connecting the foreground input.

**Foreground:** The second of two images you want to switch between or mix. The Dissolve node works best when both foreground and background inputs are connected to images with the same resolution.

**Gradient Map:** (Optional) The Gradient Map is required only when Gradient Wipe is selected.

## Basic Node Setup

Dissolve nodes are typically connected in the following way, with two input images connected to the background and foreground inputs, and the output connected to the next node in the composition.



A typical dissolve node structure in Fusion

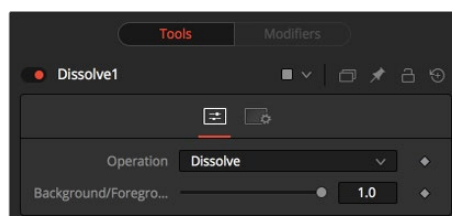
## Resolution Handling

It is recommended to make sure that all images connected to the foreground, background, and gradient map inputs of the Dissolve node have the same resolution and the same pixel aspect. This is not required, however. But, the result if you mix resolutions depends on how you set the Background/Foreground slider.

- If the input images are different sizes, but the Foreground/Background slider is set to full Foreground (all the way to the right) or full Background (all the way to the left), then the output resolution will be identical to the image resolution of the corresponding node input.
- If input images of different sizes are mixed by setting the Background/Foreground slider somewhere between, the output resolution will be set to the larger of the two input resolutions to make sure there's enough room to contain both images. In this case, you may experience undesirable resolution changes when the slider moves from full foreground or background to somewhere in between.

For example, if you try to dissolve between a 4K image (connected to the background) and an 8K image (connected to the foreground), the output of the Dissolve node will be 4K when the slider is set to full Background, but will suddenly jump to 8K when set to full Foreground or when mixed somewhere between the foreground and background.

## Inspector



Dissolve controls

## Controls Tab

These are the main controls that govern the Dissolve node's behavior.

- **Operation Pop-Up:** The Operation menu contains one of seven different methods for mixing the Foreground and Background inputs. The two images are mixed using the value of the Background/Foreground slider to determine the percentage each image contributes.
  - Dissolve:** The standard Dissolve mode is the equivalent of a cross dissolve: one clip fades out as another clip fades in.
  - Additive Dissolve:** Similar in look to a standard film dissolve, an Additive dissolve adds the second clip and then fades out the first one.
  - Erode:** The Erode method transitions between the two images by growing the darkest areas of the background image to reveal the foreground image. The effect appears similar to a filmstrip burning out.
  - Random Dissolve:** A randomly generated dot pattern is used to perform the mix of the images.
  - Random Noise Dissolve:** A moving random dot pattern is used to perform the mix of the images.
  - Gradient Wipe:** The dissolve is controlled by the luminance values of the image in the Gradient Map input. The edges of this dissolve can be softened. The density and the color of the border can be adjusted independently.
  - SMPTE Wipe:** The SMPTE wipe is similar to the basic effect wipes found on many video effects switchers. There is a horizontal wipe and a vertical wipe provided. The wipes can have soft edges and borders added. The density and the color of the border can be adjusted independently.
- **Background/Foreground Slider:** Defaults to Foreground. This control determines whether the output is the background image, the foreground image, or a mix between the two. The type of mix is determined by the Operation control. If one of the input images is not currently available, the other one will be output despite the setting of this slider.

## Gradient/SMPTE Wipe Controls

The following controls appear only when Gradient Wipe or SMPTE Wipe are selected.

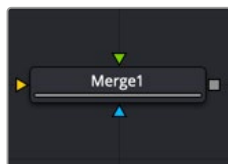
- **Wipe Style:** (SMPTE Wipe only) This drop-down list allows the selection of two wipe styles: Horizontal - Left to Right and Vertical - Top to Bottom. The direction of the wipes can be reversed by using the Invert Wipe checkbox.
- **Invert Wipe:** (SMPTE Wipe only) When checked, the direction of the wipe will be reversed.
- **Softness:** Use this control to soften the edge of the transition.
- **Border:** Select the Border to enable coloring of the transition's edge and to reveal the associated controls. The effect is to create a border around the transition edge.
- **Border Softness:** (Appears only when Border is turned on) The Border Softness slider controls the width and density of the border. Higher values will create a denser border, and lower values will create a thinner one.
- **Border Color:** (Appears only when Border is turned on) Use Border Color to select the color used in the border.

## Common Controls

### Settings Tab

The Settings tab in the Inspector is also duplicated in both the Dissolve and Merge nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

# Merge [Mrg]



The Merge node

## Merge Node Introduction

The Merge node combines two images based on the Alpha (opacity) channel associated with the one in front. This node takes two inputs: a background and a foreground image. The Operation mode determines which method is used to combine the foreground and background images, supporting the standard Over, In, Held Out, Atop, and XOr methods for compositing images. Meanwhile, an Apply Mode pop-up lets you use different composite modes, transfer modes, or blend modes (whichever is your preferred terminology) to combine the foreground against the background in different ways. This includes such standard modes as screen, dissolve, multiply, overlay, as well as many others.

The Merge node can perform both additive (premultiplied) and subtractive (non-premultiplied) compositing, depending on how your compositions and media are set up. However, you also have the flexibility of using the Additive/Subtractive slider to blend between additive and subtractive composite results, which has the bonus of providing solutions for problem edges in some cases.

Ordinarily, the foreground and background input connections determine the layer order of images composited with this node. However, you can also enable Z-Depth compositing if Z-channels are available in the input images. Z-merging compares the depth value of each pixel in each layer to determine which pixels should be in front and which should be behind.

## Inputs

The Merge node provides three image inputs, all of which are optional:

**Background:** The orange background input is for the first of two images you want to composite together. You should connect the background input before connecting the foreground input. If you connect an image to the background without connecting anything to the foreground input, the Merge node will output the background image.

**Foreground:** The green foreground input is for the second of two images you want to composite together, which is typically a foreground subject that should be in front of the background. If you connect an image to the foreground input without connecting anything to the background input first, the Merge node won't output anything.

**Effect Mask:** (Optional) The effect mask input lets you mask a limited area of the output image to be merged where the mask is white (where the foreground image shows in front of the background), letting the background image show through by itself where the mask is black.

## Basic Node Setup

Merge nodes are typically connected in the following way, with two input images connected to the background and foreground inputs, and the output connected to the next node in the composition. In this example, the effect mask input is not used, as this is not typical.



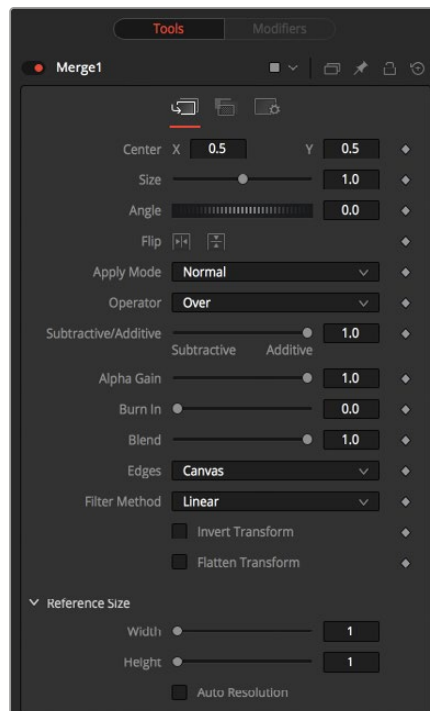
A typical Merge node structure in DaVinci Resolve

## Resolution Handling

While you can connect images of any resolution to the background and foreground inputs of the Merge node, the image that's connected to the background input determines the resolution of the output.

**TIP:** If you want to change the resolution of the image connected to the background, you can use the Crop node to change the “canvas” resolution of the image without changing the size of the original image, or you can use the Resize node to change both the resolution and the size of the image.

## Inspector



Merge node controls

## Merge Tab

The Merge tab contains most of the controls necessary for customizing most merge operations.

### Foreground Sizing Controls

These controls let you adjust the sizing of the image connected to the foreground input, making it unnecessary to use a separate Transform node to fit the foreground layer to match the background layer in simple compositions.

- **Center X and Y:** This control determines the position of the foreground image in the composite. The default is 0.5, 0.5, which centers the foreground image in the exact center of the background image. The value shown is always the actual position in normalized coordinates, multiplied by the reference size. See below for a description of the reference size controls.
- **Size:** Use this control to increase or decrease the size of the foreground image before it is composited over the background. The range of values for this slider is 0.0 to 5.0, but any value greater than 0 can be entered manually. A size of 1.0 gives a pixel-for-pixel composition, where a single pixel in the foreground is the same size as a single pixel in the background.
- **Angle:** Use this control to rotate the foreground image before it is combined with the background.

### Compositing Mode and Adjustment Controls

The next six parameters control how the background and foreground input images are combined to create a single output image.

- **Apply Modes:** The Apply Mode setting determines the math used when blending or combining the foreground and background pixels.

**Normal:** The default Normal merge mode uses the foreground's Alpha channel as a mask to determine which pixels are transparent and which are not. When this is active, another menu shows possible operations, including Over, In, Held Out, Atop, and XOr.

**Screen:** Screen merges the images based on a multiplication of their color values. The Alpha channel is ignored, and layer order becomes irrelevant. The resulting color is always lighter. Screening with black leaves the color unchanged, whereas screening with white will always produce white. This effect creates a similar look to projecting several film frames onto the same surface. When this is active, another menu shows possible operations, including Over, In, Held Out, Atop, and XOr.

**Dissolve:** Dissolve mixes two image sequences together. It uses a calculated average of the two images to perform the mixture.

**Darken:** Darken looks at the color information in each channel and selects the background or foreground image's color value, whichever is darker, as the result color. Pixels lighter than the merged colors are replaced, and pixels darker than the merged color do not change.

**Multiply:** Multiplies the values of a color channel. This will give the appearance of darkening the image as the values are scaled from 0 to 1. White has a value of 1, so the result would be the same. Gray has a value of 0.5, so the result would be a darker image, or an image half as bright.

**Color Burn:** Color Burn uses the foreground's color values to darken the background image. This is similar to the photographic dark room technique of burning by increasing the exposure of an area of a print.

**Linear Burn:** Linear Burn decreases the brightness of the base color based on the value of the blend color. The result is darker than Multiply but less saturated than Color Burn. Linear Burn also produces the most contrast in darker colors than any of the other blending modes in the Darker group.

**Darker Color:** The Darker Color blending mode is very similar to Darken. This blending mode does not blend pixels. It only compares the base and blend colors, and it keeps the darkest of the two. The difference is that Darker Color looks at the composite of all the RGB channels, whereas Darken looks at each RGB channel individually to come up with a final blend.

**Lighten:** Lighten looks at the color information in each channel and selects the background or foreground image's color values, whichever is lighter, as the result color value. Pixels darker than the merged color are replaced, and pixels lighter than the merged color do not change.

**Color Dodge:** Color Dodge uses the foreground's color values to brighten the background image. This is similar to the photographic dark room technique of dodging by reducing the exposure of an area of a print.

**Linear Dodge:** Linear Dodge (Add) produces similar but stronger results than Screen or Color Dodge. This blending mode looks at the color information in each channel and brightens the base color to reflect the blend color by increasing the brightness. Blending with black produces no change. Linear Dodge (Add) blends differently when Fill Opacity is adjusted, compared to when Opacity is adjusted.

**Lighter Color:** Lighter Color is very similar to Lighten. This blending mode does not blend pixels. It only compares the base and blend colors, and it keeps the brightest of the two. The difference is that Lighter Color looks at the composite of all the RGB channels, whereas Lighten looks at each RGB channel to come up with a final blend.

**Overlay:** Overlay multiplies or screens the color values of the foreground image, depending on the color values of the background image. Patterns or colors overlay the existing pixels while preserving the highlights and shadows of the color values of the background image. The background image is not replaced but is mixed with the foreground image to reflect the original lightness or darkness of the background image.

**Soft Light:** Soft Light darkens or lightens the foreground image, depending on the color values of the background image. The effect is similar to shining a diffused spotlight on the image.

**Hard Light:** Hard Light multiplies or screens the color values of the foreground image, depending on the color values of the background image. The effect is similar to shining a harsh spotlight on the image.

**Vivid Light:** Vivid Light is an extreme version of Overlay and Soft Light. Anything darker than 50% gray is darkened, and anything lighter than 50% gray is lightened. Vivid Light is one of those blending modes where you may want to adjust the opacity, since 100% opacity is generally too strong. Vivid Light is the fifth blending mode of eight that give you different results when you reduce the fill compared to opacity.

**Linear Light:** Linear Light uses a combination of the Linear Dodge blending on lighter pixels and a Linear Burn on darker pixels. Typically, the resulting colors are extreme, and you may want to use the Opacity or Fill sliders to adjust it. Linear Lights blends differently when Fill Opacity is adjusted, compared to when Opacity is adjusted.

**Pin Light:** Pin Light is an extreme blending mode that performs a Darken and Lighten blending mode simultaneously. It can result in patches or blotches, and it completely removes all mid-tones.

**Difference:** Difference looks at the color information in each channel and subtracts the foreground color values from the background color values or the background from the foreground, depending on which has the greater brightness value. Merging with white inverts the color. Merging with black produces no change.

**Exclusion:** Exclusion creates an effect similar to, but lower in contrast than, the Difference mode. Merging with white inverts the base color values. Merging with black produces no change.

**Hue:** Hue creates a result color with the luminance and saturation of the background color values and the hue of the foreground color values.



**Saturation:** Saturation creates a result color with the luminance and hue of the base color and the saturation of the blend color.

**Color:** Color creates a result color with the luminance of the background color value and the hue and saturation of the foreground. This preserves the gray levels in the image and is useful for coloring monochrome images.

**Luminosity:** Luminosity creates a result color with the hue and saturation of the background color values and the luminance of the foreground color values. This mode creates an inverse effect from that of the Color mode.

**Hypotenuse:** This blend mode is good for HDR images that have out of range colors above 1, It uses a square root of each color squared and added to blend the color.

$$\text{Out} = \text{sqrt} (\text{Fc}*\text{Fc} + \text{Bc}*\text{Bc})$$

**Geometric:** This blend mode is good for HDR images that have out of range colors above 1, For values above zero the result is 2 times the foreground times the background color divided by the foreground plus background color.

$$\text{Out} = 2*\text{Fc}*\text{Bc} / (\text{Fc}+\text{Bc})$$

- **Operator Modes:** This menu is used to select the Operation mode of the merge. Changing the Operation mode changes how the foreground and background are combined to produce a result. This pop-up menu is visible only when the Merge node's Apply mode is set to either Normal or Screen.

For an excellent description of the math underlying the Operation modes, read *Compositing Digital Images*, Porter, Thomas, and T. Duff, ACM SIGGRAPH Computer Graphics proceedings, 1984, pages 253-259. Essentially, the math is as described below. Note that some modes not listed in the Operator drop-down menu (Under, In, Held In, Below) are easily obtained by swapping the foreground and background inputs (with Command-T or Ctrl-T) and choosing a corresponding mode. The formula used to combine pixels in the merge is always  $(fg * x) + (bg * y)$ . The different operations determine exactly what x and y are, as shown in the description for each mode.

**The Operator modes are as follows:**

**Over:** The Over mode adds the foreground layer to the background layer by replacing the pixels in the background with the pixels from the Z wherever the foreground's Alpha channel is greater than 1.

$$x = 1, y = 1 - [\text{foreground Alpha}]$$

**In:** The In mode multiplies the Alpha channel of the background input against the pixels in the foreground. The color channels of the foreground input are ignored. Only pixels from the foreground are seen in the final output. This essentially clips the foreground using the mask from the background.

$$x = [\text{background Alpha}], y = 0$$

**Held Out:** Held Out is essentially the opposite of the In operation. The pixels in the foreground image are multiplied against the inverted Alpha channel of the background image. Accomplish exactly the same result using the In operation and a Matte Control node to invert the matte channel of the background image.

$$x = 1 - [\text{background Alpha}], y = 0$$

**Atop:** Atop places the foreground over the background only where the background has a matte.

$x = [\text{background Alpha}], y = 1 - [\text{foreground Alpha}]$

**XOr:** XOr combines the foreground with the background wherever either the foreground or the background has a matte, but never where both have a matte.

$x = 1 - [\text{background Alpha}], y = 1 - [\text{foreground Alpha}]$

**Conjoint:** The Conjoint mode will make a decision based combination of Alpha channels of the foreground and background images; this is helpful in soft edge and motion blurred Alpha where Alpha is not solid.

$X = 1, Y = X + Y(1 - af) / ab, \text{ if } af > ab$

**Disjoint:** The Disjoint mode will make a decision based combination of Alpha channels of the foreground and background images; this is helpful in combining layers as to not get out of range Alpha, and premultiplied edges get the correct Alpha combination.

$X = X + Y(1 - af) / ab, Y = X + Y \text{ if } af + ab < 1$

**Mask:** The Mask mode will output the background image multiplied by the foreground Alpha.

$X = X * af, Y = 0$

**Stencil:** The Stencil mode will output the background image multiplied by the inverse foreground Alpha.

$X = X * (1 - af), Y = 0$

**Under:** The Under mode is the same operation as the Over mode but will swap foreground and background images in the operations.

$X = Y, Y = X * (1 - af)$

- **Subtractive/Additive slider:** This slider controls whether Fusion performs an Additive merge, a Subtractive merge, or a blend of both. This slider defaults to Additive merging for most operations, assuming the input images are premultiplied (which is usually the case). If you don't understand the difference between Additive and Subtractive merging, here's a quick explanation.
  - An Additive merge is necessary when the foreground image is premultiplied, meaning that the pixels in the color channels have been multiplied by the pixels in the Alpha channel. The result is that transparent pixels are always black, since any number multiplied by 0 always equals 0. This obscures the background (by multiplying with the inverse of the foreground Alpha), and then simply adds the pixels from the foreground.
  - A Subtractive merge is necessary if the foreground image is not pre-multiplied. The compositing method is similar to an additive merge, but the foreground image is first multiplied by its own Alpha to eliminate any background pixels outside the Alpha area.

While the Additive/Subtractive option could easily have been a checkbox to select one mode or another, the Merge node lets you blend between the Additive and Subtractive versions of the merge operation—an operation that is occasionally useful for dealing with problem composites with edges that are calling attention to themselves as too bright or too dark.

For example, using Subtractive merging on a premultiplied image may lead to darker edges, whereas using Additive merging with a non-premultiplied image will cause any non-black area outside the foreground's Alpha to be added to the result, thereby lightening the edges. By blending between Additive and Subtractive, you can tweak the edge brightness to be just right for your situation.

- **Alpha Gain slider:** Alpha Gain linearly scales the values of the foreground's Alpha channel. In Subtractive merges, this controls the density of the composite, similarly to Blend. In Additive merges, this effectively reduces the amount that the background is obscured, thus brightening the overall result. In an Additive merge with Alpha Gain set to 0.0, the foreground pixels are simply added to the background.
- **Burn In slider:** The Burn In control adjusts the amount of Alpha used to darken the background, without affecting the amount of foreground added in. At 0.0, the merge behaves like a straight Alpha blend, whereas at 1.0, the foreground is effectively added onto the background (after Alpha multiplication if in Subtractive mode). This gives the effect of the foreground image brightening the background image, as with Alpha Gain. For Additive merges, increasing the Burn In gives an identical result to decreasing Alpha Gain.
- **Blend slider:** This is a cloned instance of the Blend slider in the Common Controls tab. Changes made to this control are simultaneously made to the one in the common controls. The Blend slider mixes the result of the node with its input, blending back the effect at any value less than 1.0. In this case, it will blend the background with the merged result.

### Additional Controls

The remaining controls let you fine-tune the results of the above settings.

- **Filter Method:** For input images that are being resized, this setting lets you choose the filter method used to interpolate image pixels when resizing clips. The default setting is Linear. Different settings work better for different kinds of resizing. Most of these filters are useful only when making an image larger. When shrinking images, it is common to use the Linear filter; however, the Catmull-Rom filter will apply some sharpening to the results and may be useful for preserving detail when scaling down an image.

**Nearest Neighbor:** This skips or duplicates pixels as needed. This produces the fastest but crudest results.

**Box:** This is a simple interpolation resize of the image.

**Linear:** This uses a simplistic filter, which produces relatively clean and fast results.

**Quadratic:** This filter produces a nominal result. It offers a good compromise between speed and quality.

**Cubic:** This produces better results with continuous-tone images. If the images have fine detail in them, the results may be blurrier than desired.

**Catmull-Rom:** This produces good results with continuous-tone images that are resized down. Produces sharp results with finely detailed images.

**Gaussian:** This is very similar in speed and quality to Bi-Cubic.

**Mitchell:** This is similar to Catmull-Rom but produces better results with finely detailed images. It is slower than Catmull-Rom.

**Lanczos:** This is very similar to Mitchell and Catmull-Rom but is a little cleaner and also slower.

**Sinc:** This is an advanced filter that produces very sharp, detailed results; however, it may produce visible "ringing" in some situations.

**Bessel:** This is similar to the Sinc filter but may be slightly faster.

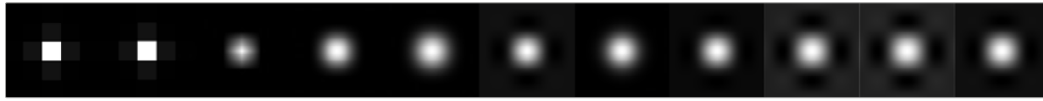
**Window Method:** Some filters, such as Sinc and Bessel, require an infinite number of pixels to calculate exactly. To speed up this operation, a windowing function is used to approximate the filter and limit the number of pixels required. This control appears when a filter that requires windowing is selected.

**Hanning:** This is a simple tapered window.

**Hamming:** Hamming is a slightly tweaked version of Hanning.

**Blackman:** A window with a more sharply tapered falloff.

**Kaiser:** A more complex window, with results between Hamming and Blackman.



Resize Filters from left to right: Nearest Neighbor, Box, Linear, Quadratic, Cubic, Catmull-Rom, Gaussian, Mitchell, Lanczos, Sinc, and Bessel

- **Edges Buttons:** Four buttons let you choose how to handle the space around images that are smaller than the current DoD of the canvas as defined by the resolution of the background image.

**Canvas:** The area outside the frame is set to the current color/opacity of the canvas. If you want to change this value, you can attach a Set Canvas Color node between the image connected to the foreground input and the foreground input itself, using Set Canvas Color to choose a color and/or transparency setting with which to fill the canvas.

**Wrap:** Creates a “video wall” effect by duplicating the foreground image as a grid.

**Duplicate:** Duplicates the outermost pixels along the edge of the foreground image, duplicating them to stretch up, down, left, and right from each side to reach the end of the DoD.

**Mirror:** Similar to duplicate, except every other iteration of the foreground image is flipped and flopped to create a repeating pattern.

- **Invert Transform:** Select the Invert Transform control to invert any position, rotation, or scaling transformation. This option is useful when connecting the merge to the position of a tracker for match moving.
- **Flatten Transform:** The Flatten Transform option prevents this node from concatenating its transformation with subsequent nodes. The node may still concatenate transforms from its input, but it will not concatenate its transformation with the node at its output.
- **Reference Size:** The controls under Reference Size do not directly affect the image. Instead, they allow you to control how Fusion represents the position of the Merge node’s center.

Normally, coordinates are represented as values between 0 and 1, where 1 is a distance equal to the full width or height of the image. This allows resolution independence, because the size of the image can be changed without having to change the value of the center.

One disadvantage to this approach is that it complicates making pixel-accurate adjustments to an image. To demonstrate, imagine an image that is 100 x 100 pixels in size. To move the center of the foreground element to the right by 5 pixels, we would change the X value of the merge center from 0.5, 0.5 to 0.55, 0.5. We know the change must be 0.05 because  $5/100 = 0.05$ .

If you specify the dimensions of the background image in the Reference Size controls, this changes the way the center control values are displayed so that it shows the actual pixel positions in its X and Y fields.

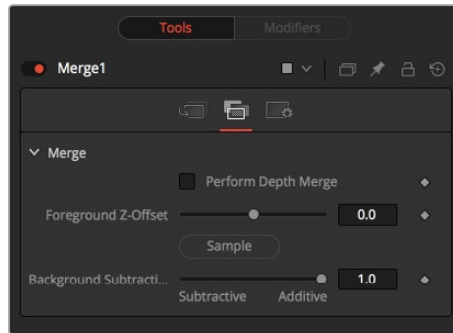
Extending the example, set the width and height to 100 each and the center will now be shown as 50, 50, and we would move it 5 pixels toward the right by entering 55, 50.

Internally, the Merge node still stores this value as a number between 0 to 1 and, if the center control’s value was to be queried via scripting or the center control was to be published for use by other nodes, the original normalized value would be retrieved. The change is only visible in the value shown for merge center in the node control.

**Use Frame Format Settings:** Select this to force the merge to use the composition’s current frame format settings to set the reference width and reference height values.

**Width and Height:** Set these sliders to the width and height of the image to change the way that Fusion displays the values of the Merge node’s center control.

## Channels Tab



Merge node Channels tab

The Channels tab has controls that let the Merge node use Z-channels embedded within each image to define what's in front and what's behind during a Merge operation. The following controls let you customize the result.

- **Perform Depth Merge:** Off by default. When turned on, the Z-channel of both images will be used to determine the composite order. Alpha channels are still used to define transparency, but the values of the Z-Depth channels will determine the ordering of image elements, front to back. If a Z-channel is not available for either image, the setting of this checkbox will be ignored, and no depth compositing will take place. If Z-Depth channels are available, turning this checkbox off disables their use within this operation.
- **Foreground Z-Offset:** This slider sets an offset applied to the foreground image's Z value. Click the Pick button to pick a value from a displayed image's Z-channel, or enter a value using the slider or input boxes. Raising the value causes the foreground image's Z-channel to be offset further away along the Z-axis, whereas lowering the value causes the foreground to move closer.
- **Subtractive/Additive:** When Z-compositing, it is possible for image pixels from the background to be composited in the foreground of the output because the Z-buffer for that pixel is closer than the Z of the foreground pixel. This slider controls whether these pixels are merged in an Additive or a Subtractive mode, in exactly the same way as the comparable slider in the Merge tab.

When merged over a background of a different color, the original background will still be visible in the semitransparent areas. An Additive merge will maintain the transparencies of the image but will add their values to the background.

## Common Controls

### Settings Tab

The Settings tab in the Inspector is also duplicated in both the Dissolve and Merge nodes. These common controls are described in the following “The Common Controls” section.

# MultiMerge [MMrg]



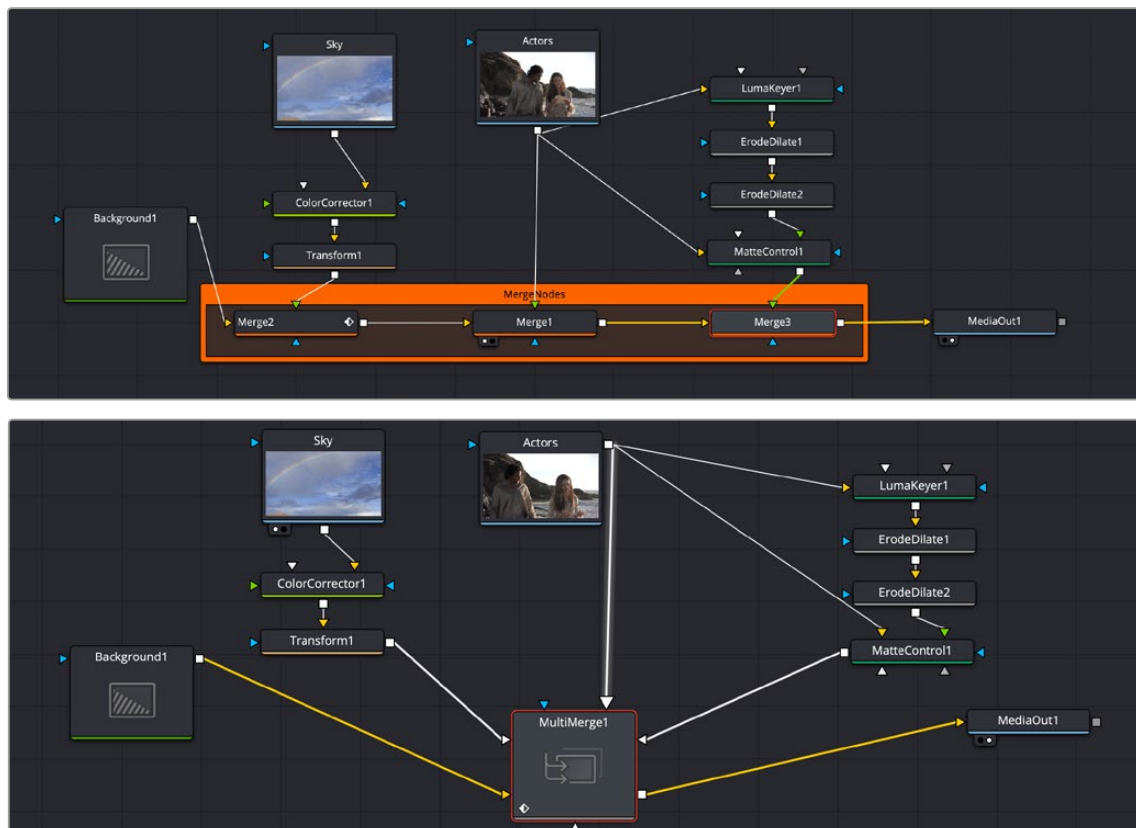
The MultiMerge node

## MultiMerge Node Introduction

The MultiMerge node allows you to combine several sources together in a layer-based structure. Unlike the standard Merge tool, which only accepts one foreground input, MultiMerge creates a new consecutive foreground input each time a new source is dragged to the tool.

These inputs are added to the Layer List in the Inspector and are displayed in a hierarchy from top to bottom as a stack. The layers are arranged so the layer closest to the foreground is at the top of the list, and the one closest to the background is at the bottom. Each layer has its own separate and independent Merge controls (see the Merge node above) that appear in the lower half of the Inspector, allowing you to individually adjust the position, size, composite modes, etc., for each source input.

As your composite increases in complexity, you can end up with a large amount of standard Merge nodes scattered throughout the node tree. MultiMerge can be used to combine these Merge nodes together, both as an organizational tool and allowing you to control the order of operations in a layer-based, rather than a node-based environment. You could conceivably composite and organize hundreds of separate layers using one MultiMerge node on your node tree.



A sky replacement composite requiring three separate Merge nodes (previous page) is replaced by one single Multi Merge node (above). The pipe of the currently selected layer in the Layer List has a slight glow applied (Actors node to MultiMerge node).

## Inputs

The MultiMerge node provides these image inputs:

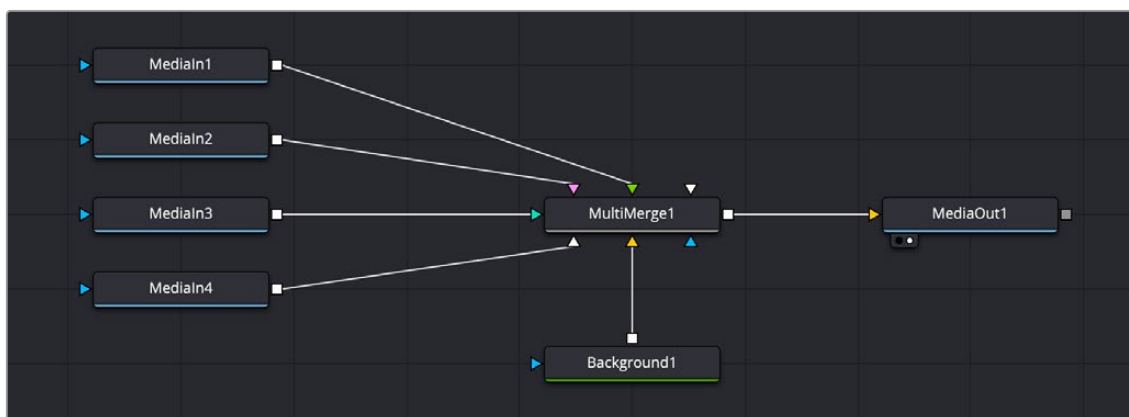
**Background:** The orange background input is for the first of the images you want to composite together. You should connect the background input before connecting the foreground input.

**Foreground:** The foreground inputs are for each subsequent image you want to composite together, which is typically a foreground subject that should be in front of the background. Connecting a pipe from a new source onto the MultiMerge node automatically creates a new foreground input on the node and a new layer in the Layer List. Foreground inputs are all colored white.

**Effect Mask:** (Optional) The effect mask input lets you mask a limited area of the output image to be merged where the mask is white (where the foreground image shows in front of the background), letting the background image show through by itself where the mask is black.

## Basic Node Setup

MultiMerge nodes are typically connected in the following way, with multiple input images connected to the foreground inputs, a background input, and the output connected to the next node in the composition. In this example, the effect mask input is not used, as this is not typical.

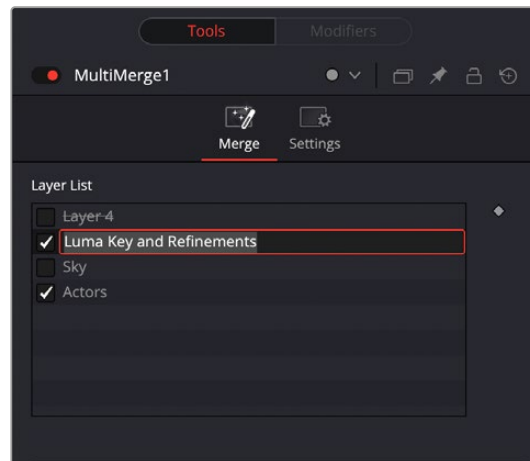


A typical MultiMerge node structure in DaVinci Resolve

## Resolution Handling

While you can connect images of any resolution to the background and foreground inputs of the MultiMerge node, the image that's connected to the background input determines the resolution of the output.

## Inspector



The Layer List of the MultiMerge node. The input to Layer 4 has been disconnected (strike through the name) and disabled (unchecked). The Luma Key layer is having its name changed, the Sky layer is disabled (unchecked), while the Actors layer is enabled (checked).

### Layer List

For each foreground input connected to the MultiMerge node, a new layer is created sequentially in the Layer List. The Layer List is hierarchically sorted from top to bottom, with layers on top being above any layers below them in the image. You can customize this layer structure in several ways.

#### To Select a Layer:

Click on the layer in the Layer List to make that layer active. The active layer's Merge controls will appear at the bottom of the Inspector, and the pipe connecting to the layer's input on the MultiMerge node will glow slightly in the node tree.

You can multi-select layers by holding the Shift key down while clicking to select a range of layers, or the Command key to select multiple non-adjacent layers.

Right-clicking on any layer opens up a contextual menu with the following options:

- **Go To Source:** Selects the tool that is connected to the layer's input on the MultiMerge node and opens the tool's parameters in the Inspector.
- **Split Here:** Automatically creates another connected MultiMerge node and splits the existing layers between the two. The layer that was selected for the split, and all layers above it, will be spun off into a new MultiMerge node. The original MultiMerge node will be connected to the background input of the new MultiMerge node.
- **Rename Layer X:** Opens up a dialog box allowing you to rename the selected layer.
- **Rename Layer to Match Source:** Automatically changes the name of the layer to the name of the tool it's connected to.
- **Set Layer X to Default:** This resets the layer's merge settings below, back to the default.
- **Delete Layer:** This deletes the layer entirely from the Layer List and disconnects the pipe from the original tool. It does not delete the original tool.



**To Move a Layer up or down the List:**

Click and drag on the Layer name and move it up or down the list. A grey line will highlight where in the list the layer will now be placed. Let go of the mouse button to make the change.

**To Rename a Layer:**

By default each layer is named based in the input tool's name. To change the Layer's name to something more meaningful, double-click on the Layer's name and type a new name in the text field. You can also right-click on a Layer's name and select Rename Layer from the contextual menu.

**To Disable/Enable a Layer:**

Checking or unchecking the box next to the Layer's name will enable or disable that layer respectively. Disabling the layer turns off that input's contribution to the overall composite in the MultiMerge but does not delete it.

**To Replace a Layer:**

By design, if you disconnect the pipe from a the MultiMerge's input, the layer will still remain in its place in the Layer List but with a strike-through its name to let you know that nothing is currently connected. This allows you to quickly iterate and audition several input sources (clips, graphics, etc.) to the same layer without having to constantly rearrange the order of your Layer List.

**Keyframing the Layer List:**

The Layer List itself is keyframable, using the standard keyframe controls to the right of the list. Parameters that can be keyframed are layer order, and layer enable and disable.

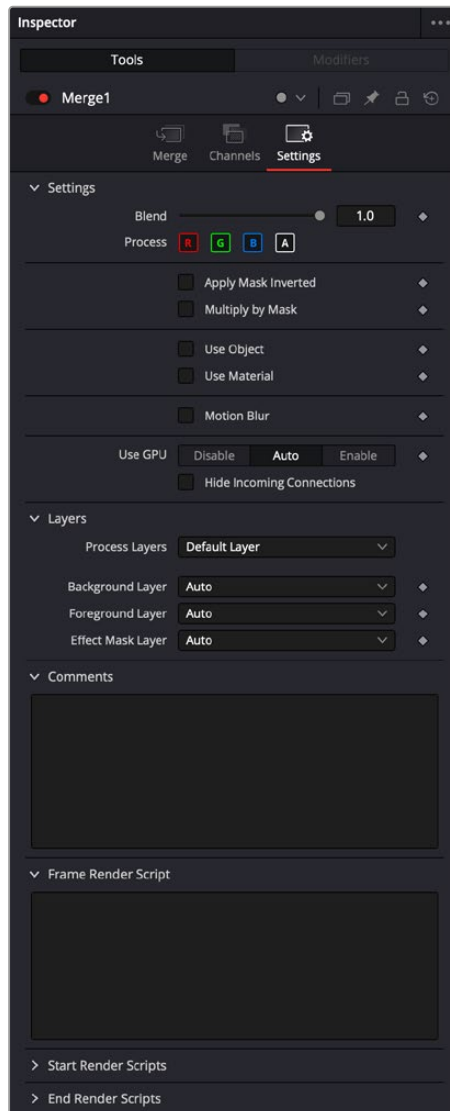
**Merge Controls**

Each layer has a separate and independent set of Merge controls that appear here when the layer is selected. For detailed information on how the Merge controls work, see the previous section in this chapter titled *Merge [Mrg]*.

# The Common Controls

The Merge, MultiMerge, and Dissolve nodes share several identical controls in the Inspector. This section describes controls that are common among those two nodes.

## Inspector



Merge Setting Inspector

## Settings Tab

The Settings tab in the Inspector can be found on both tools in the Composite category. The Settings controls are even found on third-party color type plugin tools. The controls are consistent and work the same way for each tool, although some tools do include one or two individual options that are also covered here.

### Blend

The Blend control is used to blend between the tool's original image input and the tool's final modified output image. When the blend value is 0.0, the outgoing image is identical to the incoming image. Normally, this will cause the tool to skip processing entirely, copying the input straight to the output.

### Process When Blend Is 0.0

The tool is processed even when the input value is zero. This can be useful if processing of this node is scripted to trigger another task, but the value of the node is set to 0.0.

### Red/Green/Blue/Alpha Channel Selector

These four buttons are used to limit the effect of the tool to specified color channels. This filter is often applied after the tool has been processed.

For example, if the red button on a blur tool is deselected, the blur will first be applied to the image, then the red channel from the original input will be copied back over the red channel of the result.

There are some exceptions, such as tools for which deselecting these channels causes the tool to skip processing that channel entirely. Tools that do this will generally possess a set of identical RGBA buttons on the Controls tab in the tool. In this case, the buttons in the Settings and the Controls tabs are identical.

### Apply Mask Inverted

Enabling the Apply Mask Inverted option inverts the complete mask channel for the tool. The mask channel is the combined result of all masks connected to or generated in a node.

### Multiply by Mask

Selecting this option will cause the RGB values of the masked image to be multiplied by the mask channel's values. This will cause all pixels of the image not in the mask (i.e., set to 0) to become black/transparent.

### Use Object/Use Material (Checkboxes)

Some 3D software can render to file formats that support additional channels. Notably, the EXR file format supports Object and Material ID channels, which can be used as a mask for the effect. These checkboxes determine whether the channels will be used, if present. The specific Material ID or Object ID affected is chosen using the next set of controls.

### Correct Edges

This checkbox appears only when the Use Object or Use Material checkboxes are selected. It toggles the method used to deal with overlapping edges of objects in a multi-object image. When enabled, the Coverage and Background Color channels are used to separate and improve the effect around the edge of the object. If this option disabled (or no Coverage or Background Color channels are available), aliasing may occur on the edge of the mask.

For more information on the Coverage and Background Color channels, see *Chapter 78, "Understanding Image Channels,"* in the DaVinci Resolve Reference Manual, or Chapter 18 in the Fusion Reference Manual.

### Object ID/Material ID (Sliders)

Use these sliders to select which ID will be used to create a mask from the object or material channels of an image. Use the Sample button in the same way as the Color Picker: to grab IDs from the image displayed in the view. The image or sequence must have been rendered from a 3D software package with those channels included.

## Use GPU

The Use GPU menu has three settings. Setting the menu to Disable turns off hardware-accelerated rendering using the graphics card in your computer. Enabled uses the hardware. Auto uses a capable GPU if one is available and falls back to software rendering when a capable GPU is not available

## Motion Blur

- **Motion Blur:** This toggles the rendering of Motion Blur on the tool. When this control is toggled on, the tool's predicted motion is used to produce the motion blur caused by the virtual camera's shutter. When the control is toggled off, no motion blur is created.
- **Quality:** Quality determines the number of samples used to create the blur. A quality setting of 2 will cause Fusion to create two samples to either side of an object's actual motion. Larger values produce smoother results but increase the render time.
- **Shutter Angle:** Shutter Angle controls the angle of the virtual shutter used to produce the motion blur effect. Larger angles create more blur but increase the render times. A value of 360 is the equivalent of having the shutter open for one whole frame exposure. Higher values are possible and can be used to create interesting effects.
- **Center Bias:** Center Bias modifies the position of the center of the motion blur. This allows the creation of motion trail effects.
- **Sample Spread:** Adjusting this control modifies the weighting given to each sample. This affects the brightness of the samples.

## Layers

- **Process Layers:** Select the layer to process for effects from:
  - Default Layer (default) - To apply the effect on the base/unnamed layer, not affecting other layers.
  - All Layers - To apply the effect on all available layers.
  - Custom - Choice of other available layers to apply the effect on, not affecting other layers.
- **Input Layer:** The Input Layer lists all layers available from the upstream node of the default input. The selected layer can be used by the current tool or farther down the node tree in isolation.
- **Effect Mask Layer:** If the tool has an Effect Mask input, this control lists the available layers that can provide mattes for the tool.

## Comments

The Comments field is used to add notes to a tool. Click in the field and type the text. When a note is added to a tool, a small red square appears in the lower left corner of the node when the full tile is displayed or a small text bubble icon appears on the right when nodes are collapsed. To see the note in the node editor, hold the mouse pointer over the node to display the tooltip.

## Scripts

Three Scripting fields are available on every tool in Fusion from the Settings tab. They each contain edit boxes used to add scripts that process when the tool is rendering. For more details on scripting nodes, please consult the Fusion scripting documentation.